

Xpat — Nairobi urban flood model

Submission note, Team A: data sources, assumptions and the AI feature, with how each part of the brief is met. Generated 2026-10-08 from the model itself (config nairobi-prototype-v0.6, fingerprint acedb441ec75).

Measure (synthetic starter portfolio)	Value
Properties · insured value	600 · KES 63.64 bn (SYNTHETIC)
1-in-100 ground-up loss (scenario)	KES 1.70 bn (2.67% of value)
1-in-250 ground-up loss (scenario)	KES 2.65 bn (4.16% of value)
Average annual loss: ground-up · gross · reinsurance recoveries · net	KES 125.9 m · KES 114.8 m · KES 44.9 m · KES 69.9 m
Named flood areas flagged by the hazard proxy	12 of 24 (the brief's known limit)
AI free-text ingestion, held-out cases fully correct	12 of 12

1. How it works

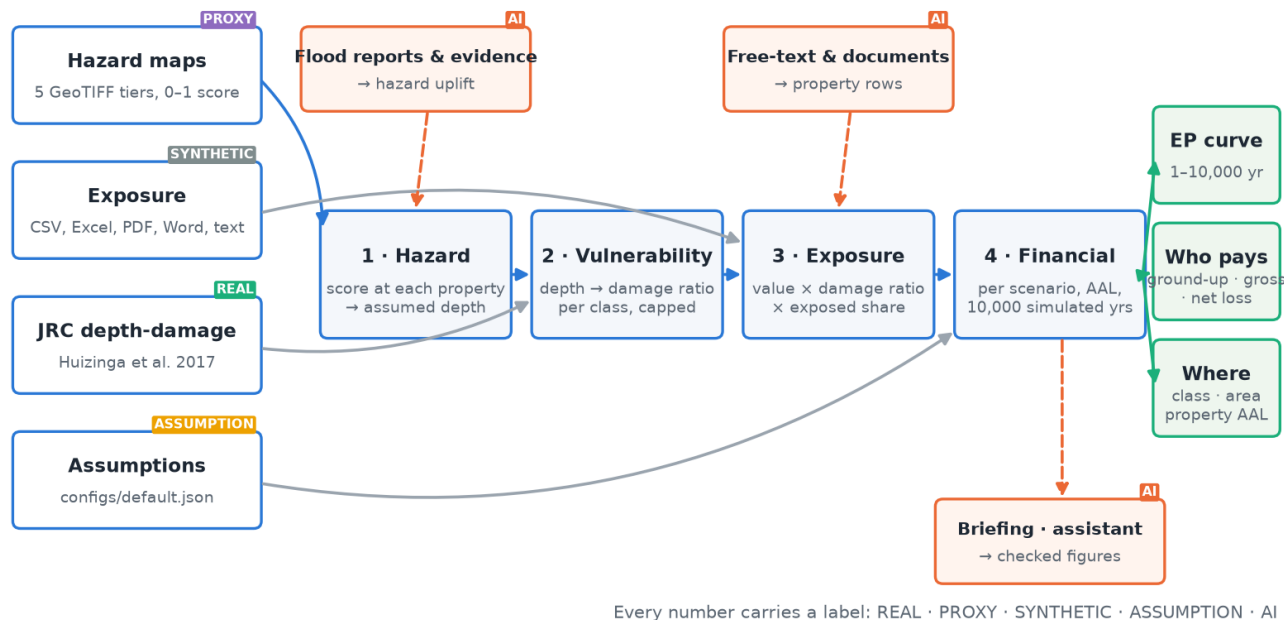


Diagram 1. The four-stage pipeline the brief prescribes (hazard → vulnerability → exposure → financial engine), its inputs with their labels, and where the AI enters (orange, dashed).

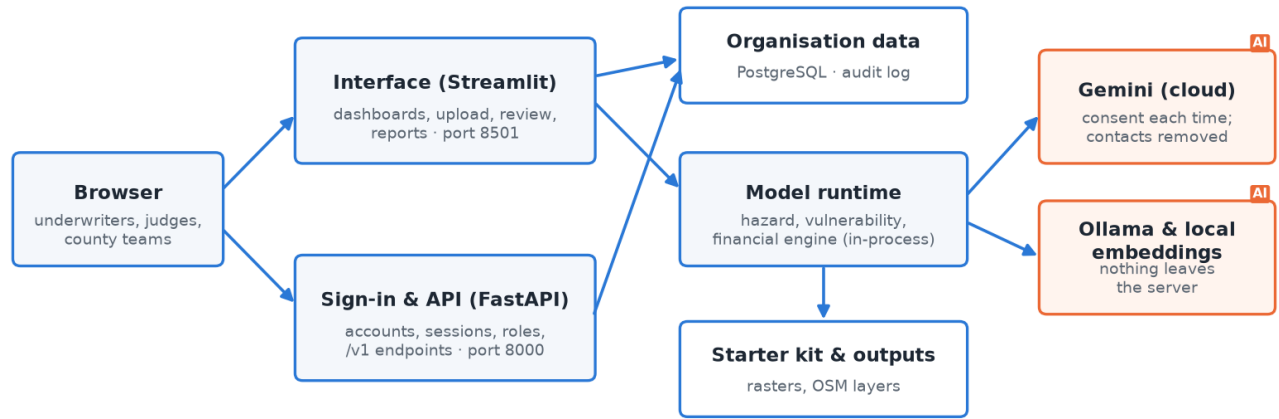


Diagram 2. The running system: a web interface and a sign-in/API server over one in-process model; organisation data and the audit log in PostgreSQL; AI through Gemini (with consent) or models on the server.

2. Data sources

Data	Use	Label
Five pluvial proxy rasters (starter kit): terrain elevation, depressions, slope (Copernicus GLO-30) and distance to OSM rivers	Hazard score at each property, per tier	PROXY
24 geocoded government-named flood hotspots (county list, March 2026; OSM Nominatim)	Validation and accumulation tags only, never exposure or training	REAL (names), approximate points
600-building starter portfolio (starter kit)	Exposure for every published figure	SYNTHETIC
Huizinga, de Moel & Szewczyk (2017), JRC105688, Table 3-1, Africa residential	Base depth-damage curve	REAL
OpenStreetMap (Geofabrik Kenya extract, Oct 2026; ODbL): buildings, places, drains	Drainage model layers, place names in reports, infrastructure check	REAL / PROXY
Flood reports: ReliefWeb (public), uploads, GDELT news	Drainage evidence after human review	AI-read, reviewed
Uploads by users: schedules and broker documents	Exposure when real data is used	REAL or SYNTHETIC, declared

3. Hazard: five tiers and what they mean

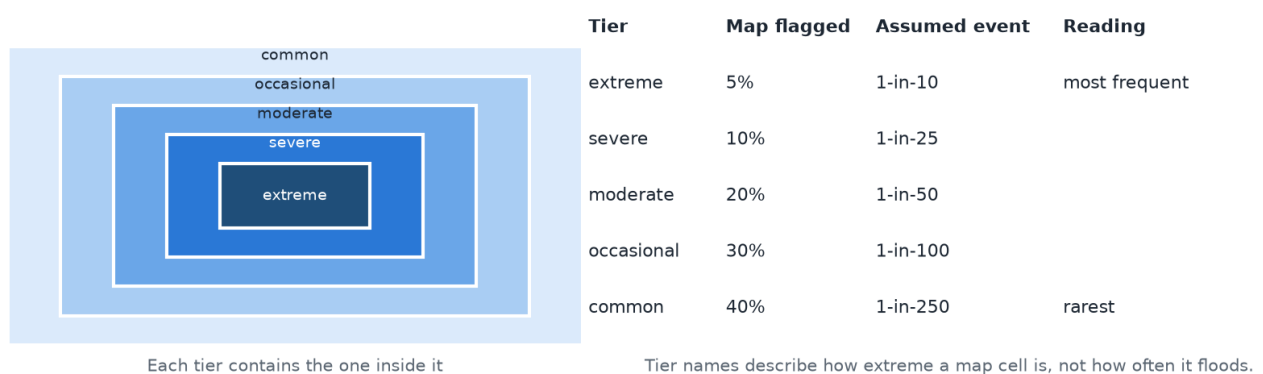


Diagram 3. The five tiers are nested threshold cuts of one susceptibility surface. A frequent flood reaches only the most susceptible cells (extreme, 5%); a rare one reaches the widest area (common, 40%). Return periods are ASSUMED.

Score to depth (ASSUMPTION). The score is relative susceptibility, not depth. We use depth = score $\times$ 1.5 m — a Team A judgement of street and household flood depth, not a cited figure, and the largest single lever on every loss (4 m multiplies losses by about 2.4). The brief's 4 m example is kept as a sensitivity case.

4. Vulnerability: a sourced curve, adapted transparently

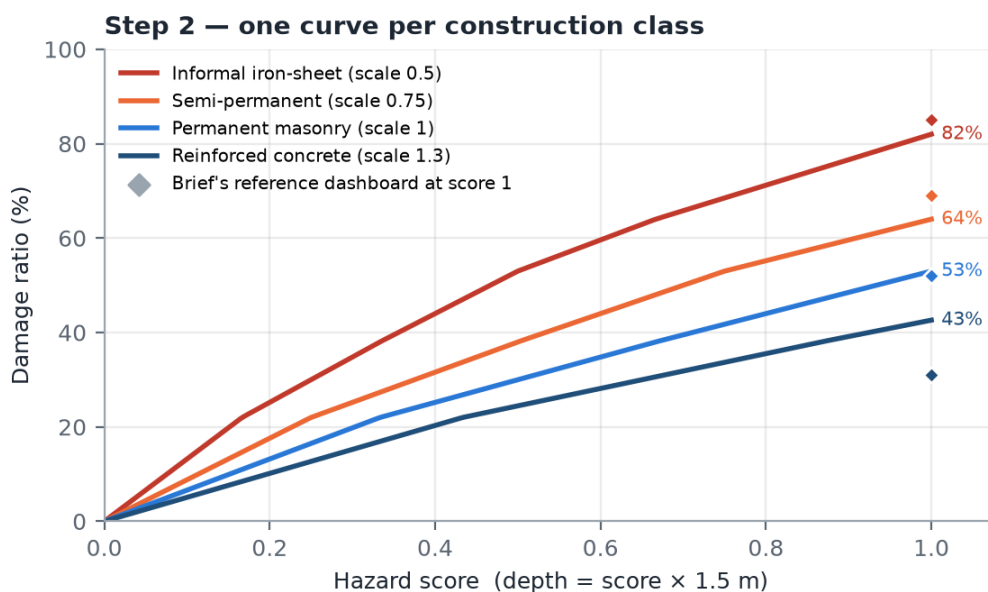


Figure 1. Damage against hazard score per construction class: the JRC Africa residential curve read at depth ÷ class scale (0.5 / 0.75 / 1.0 / 1.3), capped at 95 / 90 / 85 / 80%. Diamonds: the brief's reference dashboard.

Mean damage ratio of flagged properties	extreme (1-in-10)	severe (1-in-25)	moderate (1-in-50)	occasional (1-in-100)	common (1-in-250)
Informal iron-sheet	24.8%	28.7%	19.8%	19.8%	21.1%
Semi-permanent	14.9%	14.0%	13.9%	14.4%	14.4%
Permanent masonry	15.4%	17.9%	12.5%	13.4%	12.9%
Reinforced concrete	14.0%	8.5%	9.6%	8.4%	8.4%

The damage tiers by severity category that §9 Step 2 asks for: the average damage ratio of the properties each tier flags, by class, from the same curves. The full vulnerability matrix (class × score) is on the Assumptions page.

5. Exposure and the financial engine

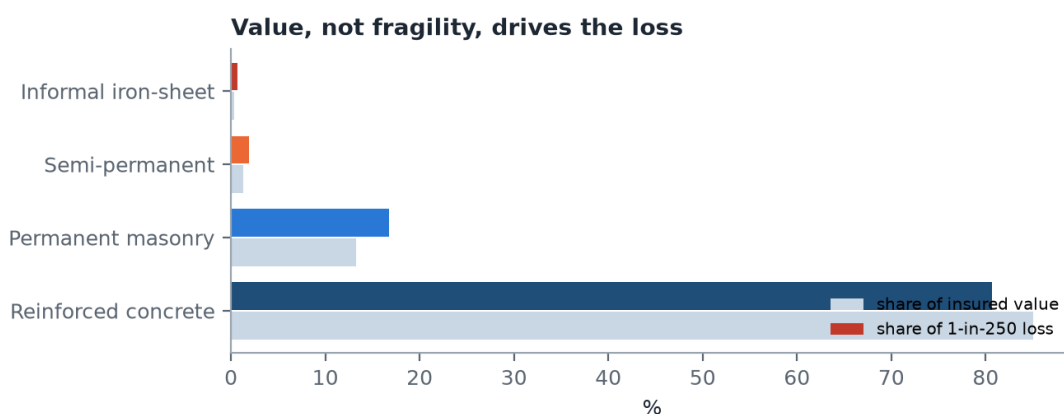


Figure 2. Reinforced concrete holds most of the insured value and most of the loss, even though informal iron-sheet housing is the most fragile class.

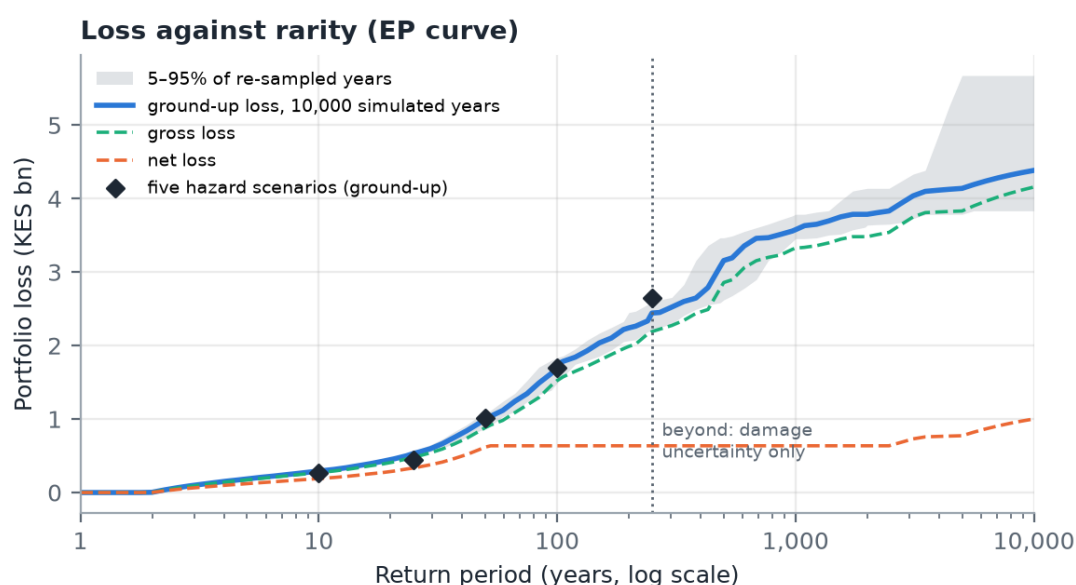


Figure 3. The loss/return-period (EP) curve: five scenario points and 10,000 simulated years with their re-sampling band; gross-loss and net-loss curves dashed. Beyond 1-in-250 no rarer flood is modelled.

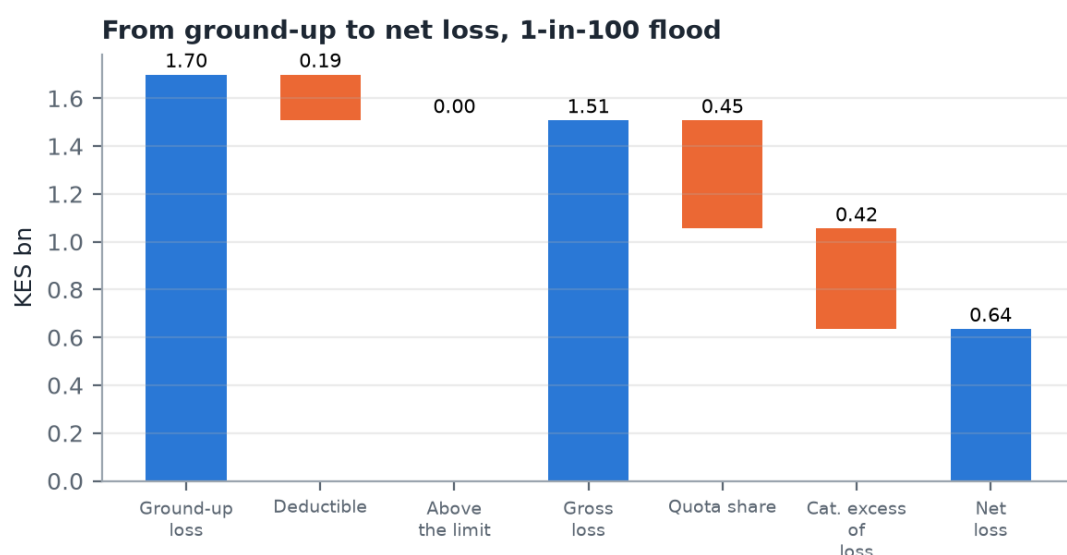


Figure 4. The 1-in-100 flood from ground-up to net loss: owners bear the deductible (1%) and anything above the limit; the quota share (30%) and the catastrophe excess of loss reduce the gross loss to the net loss. Terms and programme are illustrative ASSUMPTIONS.

Term	Meaning
Ground-up loss	The total physical damage caused to a building before insurance rules are applied.
Deductible	The part of the loss that the building owner must pay themselves before the insurer pays.
Limit	The maximum amount the insurer will pay for the building.
Gross loss	The amount the insurer is responsible for paying after applying the deductible and limit.
Quota share	A reinsurance arrangement where the insurer and reinsurer share every loss by an agreed percentage. For example, a 25% quota share means the reinsurer pays 25% of the gross loss.
Catastrophe excess of loss	Reinsurance that protects the insurer when the total loss from one catastrophe becomes large. The reinsurer starts paying after the loss passes an agreed threshold and pays up to an agreed maximum.
Net loss	The amount of the loss that remains with the insurer after payments from reinsurers.

Return period	Ground-up	Deductible	Gross loss	Quota share	Cat. XL	Net loss
1-in-10	KES 263.8 m	KES 16.3 m	KES 247.5 m	KES 74.2 m	KES 0 k	KES 173.2 m

Return period	Ground-up	Deductible	Gross loss	Quota share	Cat. XL	Net loss
1-in-25	KES 447.2 m	KES 48.7 m	KES 398.5 m	KES 119.5 m	KES 0 k	KES 278.9 m
1-in-50	KES 1.01 bn	KES 117.9 m	KES 894.0 m	KES 268.2 m	KES 0 k	KES 625.8 m
1-in-100	KES 1.70 bn	KES 191.0 m	KES 1.51 bn	KES 452.6 m	KES 419.6 m	KES 636.4 m
1-in-250	KES 2.65 bn	KES 297.3 m	KES 2.35 bn	KES 705.6 m	KES 1.01 bn	KES 636.4 m
Average annual loss	KES 125.9 m		KES 114.8 m			KES 69.9 m

6. The AI feature, and evidence that it changes the output

The brief asks for AI that **materially changes** the result, with evidence of what it contributed. Two routes do, and both are measured below; a third (the briefing and assistant) explains results with every figure checked against the model.

A • Free-text and document ingestion. A plain-English description or a broker document becomes validated property rows (same schema as a CSV), so the AI decides what is modelled. Held-out example: “Portfolio: 30 mabati structures in Kiambu (150k each); 4 stone houses in Donholm (3.5m each); 2 concrete apartment blocks in South B (80m each).” → gemini-3.5-flash produced 36 rows (30 × informal_iron_sheet in Kiambu; 4 × permanent_masonry in Donholm; 2 × concrete_rcc in South B), every quote verified in the text; modelled 1-in-250 loss KES 1.3 m. On the 12 held-out cases: 12 fully correct.

Reading a whole submission. The same single AI call also reads the document as a whole, and fixed rules then: (1) **locate data points across the text** — sums insured, premium, shares, deductions, period, parties, deductibles, extensions — each kept only if its quote and number are found word-for-word; (2) **separate relevant from irrelevant** clauses, with a reason (a bush-fire clause is set aside; a debris-removal extension is kept because it adds to a loss); (3) **cross-reference the building with risk indicators** — construction against the damage curve, occupancy against the residential curve, the location against the hazard map and the named flood areas; (4) **summarise flood history** as a dated timeline, or say there is none and what to ask the broker; (5) **calculate implied metrics** — premium rate and per mille, premium after deductions, share arithmetic checks, and modelled loss against premium once the model has run. The premium and share then pre-fill the underwriting decision.

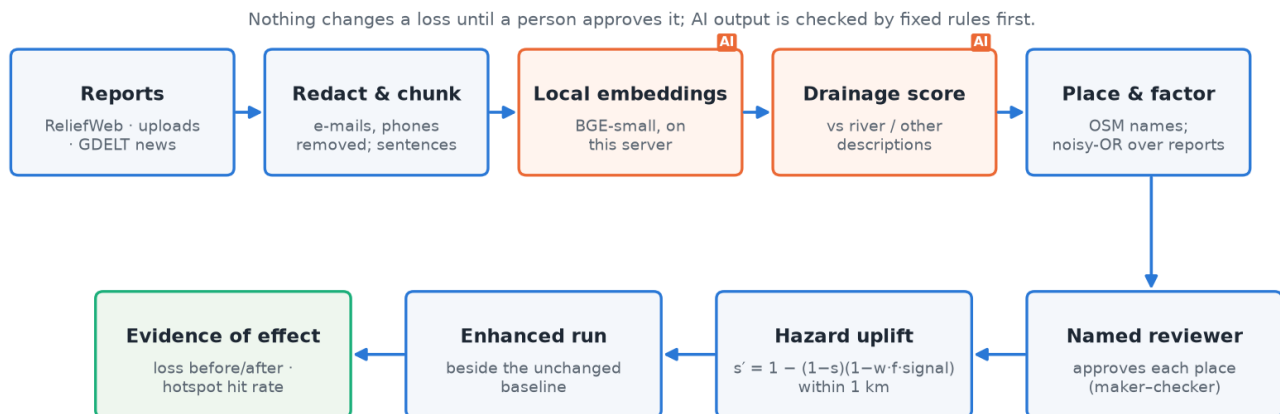


Diagram 4. The drainage-evidence loop: flood reports are read on the server by a local embedding model; places get a drainage-deficit factor; a named reviewer approves; approved evidence raises hazard nearby.

B • Flood reports → drainage evidence. Three DEMONSTRATION reports (SYNTHETIC, approval simulated) about places away from the named hotspots: the local model found Githogoro (0.96), Marurui (0.96), Kaloleni (0.88), Kabiria (0.70). Applied to the starter portfolio, 21 properties' hazard rose and the average annual loss moved from KES 125.9 m to KES 139.6 m.

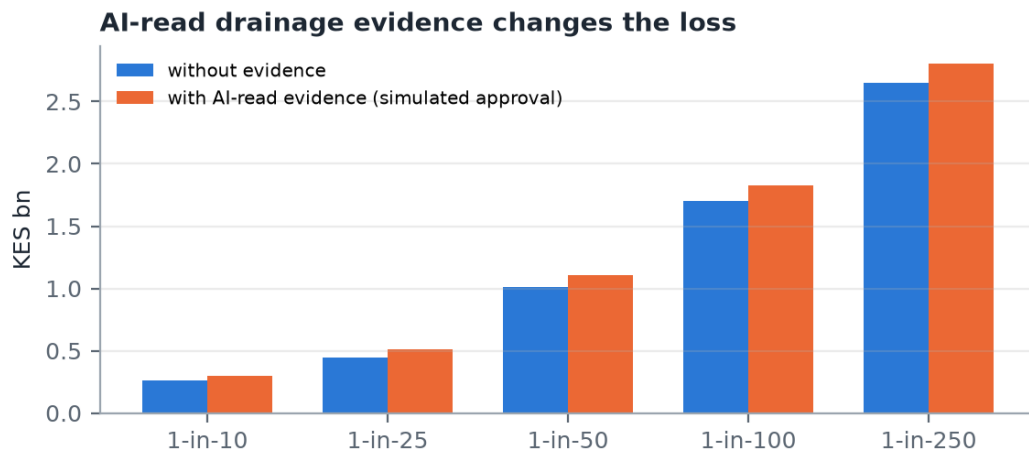


Figure 5. Portfolio loss per scenario without and with the AI-read evidence. A higher loss is not proof of a better model; the named-hotspot hit rate (independent evidence only) is the check.

AI check	Result	What it shows
Free-text ingestion, 12 held-out descriptions	12/12 fully correct	Extraction of class, count, value, place; quotes verified
Drainage-passage scoring, held-out passages	precision 100%, recall 83%	Separates drainage failure from river overflow (0 rivers flagged)
Building-density check vs 24 named areas	12 → 21 of 24	What density alone explains; 3 marginal; weak evidence

Safeguards: AI output is untrusted data and is re-validated by fixed rules; it never sets a depth, damage ratio or loss directly; contact details are removed before any AI call and users consent before text goes to Gemini; nothing reaches losses until a named reviewer approves.

7. The interface

Stakeholder	Where to look	What they see first
Underwriters & risk analysts	Overview · Loss curve · Underwriting decision	Loss at 1-in-10...10,000, AAL, who pays, accept / share / decline advice
Portfolio & exposure managers	Accumulation map · Property explorer	Value and loss by area and property, each property's AAL
Judges	Data & honesty · Methods · Assumptions	Real vs assumed, the 12/24 check, every assumption editable
County & disaster bodies	Public risk notes · Hazard checks	Plain English and Kiswahili notes, drainage and satellite checks
Cedants & brokers	Portfolio upload	CSV, Excel, PDF, Word or a description → results in one flow

The minimum the brief asks for is on the first screen: total exposure, losses at key return periods, the EP curve, the breakdown by construction class and the AI output. Every chart carries what it shows, how to read it and where it comes from, with labels.

8. Main assumptions

Assumption	Value	Basis
Tier → return period	10 / 25 / 50 / 100 / 250 years	Direction from nested footprints; years assumed
Score → depth	score × 1.5 m	Team judgement; biggest lever
Class depth scales · caps	0.5 / 0.75 / 1.0 / 1.3 · 95 / 90 / 85 / 80%	Construction reasoning · brief's 80–95% guidance
Deductible · limit	1% · 100% of value per property	Illustrative policy terms
Quota share · catastrophe XL	30% · above 1% up to 3% of value per catastrophe	Illustrative programme, not a treaty
AAL	zero below 1-in-2; rarest loss held beyond 1-in-250	Stated integration choices
Uncertainty	damage σ 0.4, correlation 0.5; 200 bootstraps of 10,000 years	Judgement; not confidence intervals

Full register (A1–A22) in docs/REPORT.md §12; every value lives in configs/default.json and is editable on the Assumptions page.

9. The brief, section by section

Brief	Requirement	Status	Where
§2.1	End-to-end pipeline to insured/reinsured loss and EP curve	Met	Diagram 1; §5 (ground-up → gross → net); Overview
§2.2	AI that materially enhances the result	Met	§6 A and B, measured
§2.3	EP curve readable by a non-modeller	Met	Figure 3; tooltips in words
§2.4	Every assumption and synthetic use stated	Met	§8; labels on every number; Data & honesty
§2.5	Interface any stakeholder can use	Met	§7; one-form sign-up, demo accounts
§3	Designed for the five stakeholder groups	Met	§7 table
§5	Ingest hazard; sourced curve; synthetic run → loss and curve; AI; real vs assumed	Met	§3–§6
§6	Synthetic exposure labelled; curve source and differences; proxy limits	Met	§2–§4; 12/24 stated
§7 in scope	Hazard, vulnerability, exposure, loss engine, one AI stage, interface	Met	All sections
§7 out of scope	Treaty structuring, layers, net-of-reinsurance loss	Exceeded (by choice)	Illustrative quota share + layer for Objective 1's "reinsured" loss; labelled, switchable
§9 Steps 1–4	Severity dataset; matrix and tiers; structured exposure; scenario losses and EP	Met	§3–§5
§9 Step 5	AI with evidence of contribution	Met	§6; outputs/ai_effect.md
§9 Step 6	Exposure, key losses, EP, class breakdown, AI output on one screen	Met	Overview
§10	Raster lookups, per-class curves, Monte Carlo year-loss simulation	Met	rasterio lookup; 10,000-year YLT
§11	Working demo and a short note on data, assumptions and AI	Met	make app; this note

10. Limitations

- The hazard is a terrain-and-river proxy, not a flood model: it flags 12 of 24 named areas and cannot see drainage on its own.
- Return periods and the score-to-depth conversion are assumptions; the curve compares assumed scenarios, not a calibrated forecast.

- The damage curve is regional (South Africa, Mozambique) and adapted by judgement; nothing is calibrated to Kenyan claims.
- The AI-evidence demonstration uses invented reports with simulated approval; real reports and a real reviewer are needed for a claim of skill.
- Policy terms and the reinsurance programme are illustrative; no reinstatements, aggregate covers or second events in a year.

Reproduce: make app · make outputs · make eval-ingestion · make eval-drainage · make eval-imd · scripts/demo_ai_effect.py · scripts/build_submission_report.py. Detail: docs/REPORT.md.